

Mock exam 1

Dmytro Shytikov

In vitro fertilization (IVF) and pregnancy tests

A girl underwent IVF, and she wants to know whether she is pregnant or not. From the previous studies, we know that the chance of conception after IVF is 0.222 (CDC, 2021).

She used an early pregnancy detection kit that gives the correct answer 95% time if the person is pregnant and 80% time if the person *is not* pregnant. That is: it detects the elevated level of hCG in 95% of cases and does not react below the threshold value in 80% of cases. Answer the questions below and provide clear calculations.

Questions

1. What is the chance that a girl with a positive test is really pregnant? Is this result convincing? (10 points.)
2. What is the chance that a girl with a *negative* test is still pregnant? (3 points.)
3. She decided to be double sure. After some time, she used the same test once again, and she got a positive result *once again*. What would be the chance that she is pregnant given she got positive results *twice*? (5 points.)
4. What can you say about this test kit? Is it good? Is it specific? Is it sensitive? Support your claims with some calculations. **Note:** use odds and odds ratios to support your words. (7 points.)

Originally created by Dmytro Shytikov in 2024.